Graph-Based Dependency Parsing

Dependency Parsing: Main Approaches

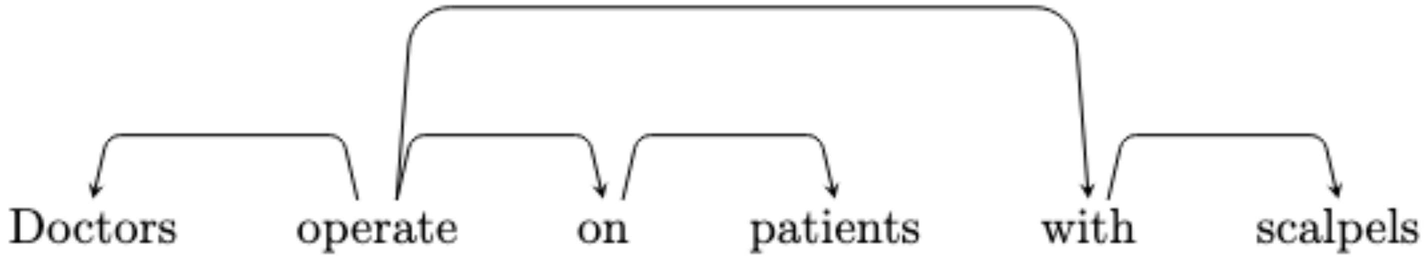
- Dynamic programming:
  - Based on context-free grammars
- Transition-based algorithms:
  - Greedy
  - Left-to-right traversal of the text, each choice is done with a classifier
- Graph algorithms:
  - Structured prediction
  - The sentence is given in advance

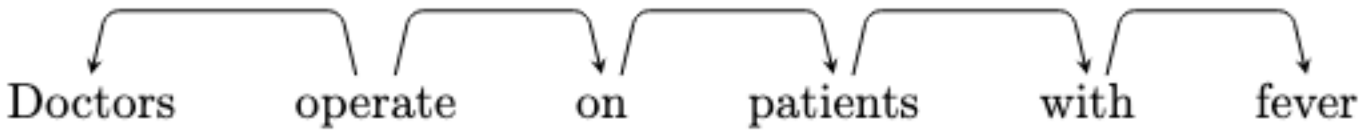
Covered in this course

Dependency Parsing

- What are the sources of information for dependency parsing?
  - Bi-lexical affinities:

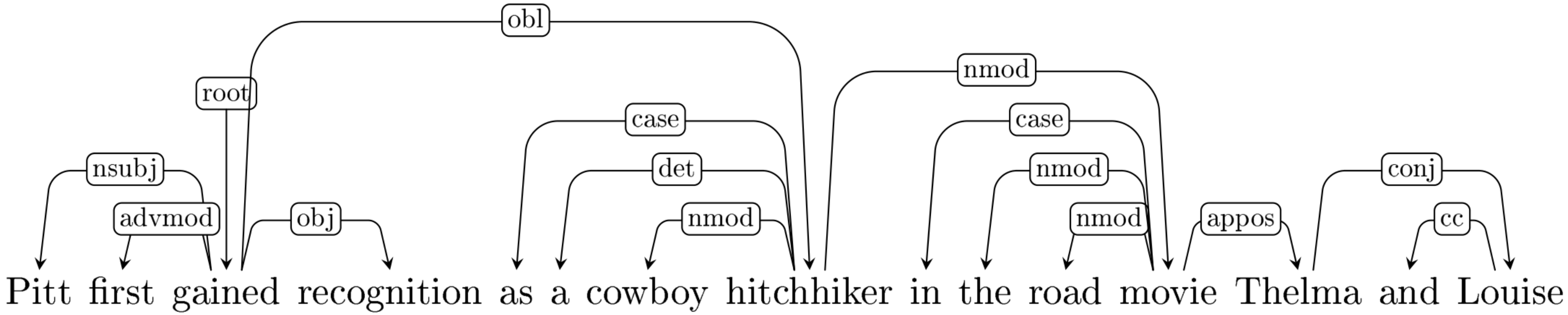
    [operate → Doctors] is plausible, [patients → scalpels] less so





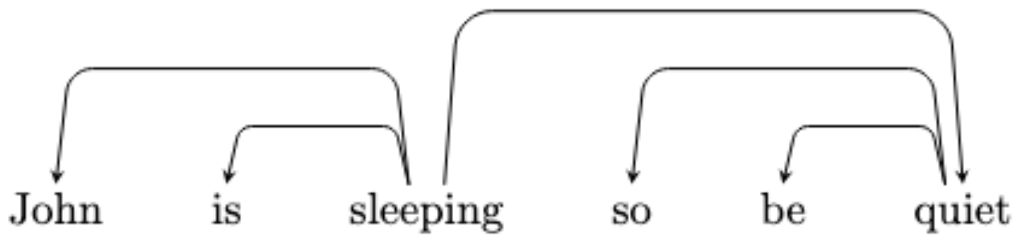
Dependency Parsing

- What are the sources of information for dependency parsing?
  - Bi-lexical affinities:
    [operate → Doctors] is plausible, [patients → scalpels] less so
  - Dependency distance:
    dependencies are more likely to be with nearby words
  - Intervening tokens:
    dependencies rarely span intervening verbs or punctuation



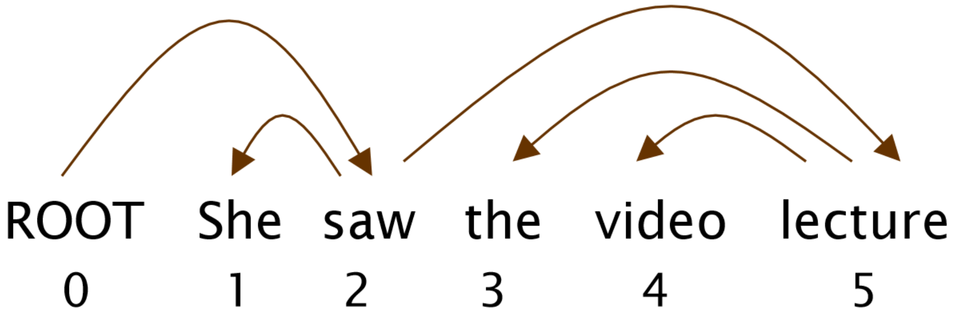
Dependency Parsing

- What are the sources of information for dependency parsing?
  - Valency: how many dependents on which side are usual for a head?
    - [gained] usually has a subject and an object
    - [sleep] does not have an object



    - [give] often has two objects (e.g., ``give John a present'')

Dependency Parsing Evaluation



$$ACC = \frac{\#CORRECT\ EDGES}{\#NUMBER\ OF\ WORDS}$$

- Accuracy (aka Attachment Score)
- There is Unlabeled Attachment Score and Labeled Attachment Score

Gold

1	2	She	nsubj
2	0	saw	root
3	5	the	det
4	5	video	nn
5	2	lecture	dobj

Parsed

index	head index	word	edge label
1	2	She	nsubj
2	0	saw	root
3	4	the	det
4	5	video	nsubj
5	2	lecture	ccomp